Week 3 · Class 1: Practice Exercises

Experimental Design in Political Science

2025-12-31

1 Segment A — Experimental Design in Political Science

2 Non-AI Exercises

2.1 1. Experimental Design Concepts

2.1.1 1.1 Multiple Choice: Experimental Validity

Internal validity refers to:

a) How well results generalize to other populations

b) Whether the treatment actually caused the observed effect

c) How large the treatment effect is

d) Whether the experiment was conducted ethically

Answer: **B**_

2.1.2 1.2 Fill-in-the-Blank: Types of Experiments

Complete the following definitions:

a) A _______________ experiment tests multiple factors simultaneously to examine both main effects and interactions.

b) A _______________ experiment manipulates scenarios or stories that participants read.

c) A _______________ experiment asks participants to choose between profiles with randomly varied attributes.

d) A _______________ experiment takes place in real-world settings with policy interventions.

Answer: a) **factorial_** b) ***vignette*** *c)* ***cojoint*** d) **field_**

2.1.3 1.3 Matching: Experimental Concepts

Match each concept with its correct definition:

Concepts:

1. Hawthorne Effect
2. SUTVA
3. Manipulation Check
4. Attrition

Definitions:

a) Participants changing behavior because they know they're being studied

b) Question verifying participants understood the treatment

c) Assumption that one person's treatment doesn't affect another's outcome

d) Participants dropping out of the study

Answer: 1-___a_ 2-___c_ 3- b 4-___d_

2.1.4 1.4 Matching: Implementation Threats

Match each concept with its correct definition:

Concepts:

1. Demand effect
2. Attention check
3. Differential attrition
4. Intent-to-treat (ITT)

Definitions:

a) An item that catches respondents (or bots) who are not actually reading the survey
b) Comparing outcomes by assigned condition, regardless of whether participants complied
c) Participants guessing the hypothesis and adjusting their answers accordingly
d) Dropout rates differing between treatment and control, breaking comparability

Answer: 1-___c_ 2-___a_ 3-___b_ 4-___d_

2.1.5 1.5 True/False: Pre-Analysis Plans

Mark each statement as True (T) or False (F):

a) Pre-analysis plans should be written after seeing the data _____
b) PAPs help prevent p-hacking and selective reporting _____
c) You cannot deviate from a PAP under any circumstances _____
d) PAPs make null results more publishable _____

Answer: a) f b) t c) f d) t

2.2 2. Designing a Study: What to Vary and What to Measure

2.2.1 2.1 Identify the Manipulation (X) and Outcome (Y)

For each study, name what was varied (X) and what was measured (Y):

a) Tomz and Weeks told respondents about a country building nuclear weapons and randomly described it as a democracy or an autocracy, then asked whether the U.S. should strike.

b) Gerber, Green, and Larimer mailed voters messages with increasing social pressure, then checked the official record of who turned out.

c) Butler and Broockman emailed state legislators from a putatively Black or White constituent asking for help registering to vote, and recorded who replied.

Answer: a) X: **the story told_** Y: _ the respondent's decision to strike or not______ b) X: **voters mesages_** Y: **voter turnout_** c) X: **email_** Y: **_replied or not**

2.2.2 2.2 The Bundle Problem

A student designs an experiment where the treatment ad uses ominous music, dark imagery, and an attack script, while the control ad uses upbeat music, bright imagery, and a positive script. The treatment ad lowers candidate support. Why can't the student conclude that "negative content" caused the drop, and what should they change?

Answer: Both variables have conflicting effects on each other; They should instead give each variable to separate respondents

2.2.3 2.3 Choosing a Control

A researcher wants to test whether the *content* of a get-out-the-vote mailer increases turnout. The treatment group receives the mailer. For the cleanest test of content, the control group should receive:

a) Nothing at all
b) A mailer of the same length and format about an unrelated civic topic (a placebo)
c) A phone call instead of a mailer
d) The same mailer one week later

Answer: _**B**

2.2.4 2.4 Behavioral vs. Attitudinal Outcomes

Label each outcome as behavioral (B) or attitudinal (A):

a) A 1–7 scale of how warmly you feel toward a candidate _____
b) Whether the participant clicked a link to donate _____
c) Whether a registered voter actually turned out, from the voter file _____
d) Agreement with the statement "immigration helps the economy" _____

Answer: a) A b) B c) B d) A

2.3 3. Implementation Threats

2.3.1 3.1 Intent-to-Treat vs. Treatment-on-Treated

A canvassing experiment assigns 1,000 voters to be contacted and 1,000 to control. Only 600 of the treatment group answer the door. Turnout is 40% among everyone assigned to treatment and 34% among controls.

a) What is the intent-to-treat (ITT) effect?

b) Will the treatment-on-treated effect (the effect among those actually canvassed) be larger or smaller than the ITT? Why?
c) Which should the researcher report as the headline, and why?

Answer: 1. ITT = 40% − 34% = 6% 2. Larger. The ITT spreads the effect across everyone assigned, including the 400 never reached; concentrating the same total effect among the ~600 compliers gives a bigger per-person effec 3. The ITT results because it answers the question much more directly

2.3.2 3.2 Differential Attrition

An experiment's treatment is a long, upsetting video; the control is a short, pleasant one. More people quit the survey in the treatment group. Why is this differential attrition more dangerous than dropout that affects both groups equally?

Answer: The results are essentially null because the samples from the treatment group are significantly smaller than then control group

2.3.3 3.3 Manipulation-Check Placement

Where should you place a manipulation check that asks "Did the article you read describe the candidate as a Democrat or a Republican?"

a) Before the outcome question, to confirm the treatment landed first
b) After the outcome question, so it does not re-prime the treatment or signal the hypothesis
c) It does not matter where it goes
d) It should replace the outcome question

Answer: **B_**

2.3.4 3.4 Statistical Power

Which statement about underpowered (too-small) experiments is correct?

a) They are only a problem because they miss real effects
b) When they do reach statistical significance, the estimated effect tends to be overstated
c) A significant result from a small study is more trustworthy than one from a large study
d) Power depends only on sample size, not on the effect size or the noise in the outcome

Answer: **_B**

3 AI Exercises

Tips for Working with Claude:

- Ask for **R code using only tidyverse** (no other packages)
- Request **simple, focused answers** to your specific question—not complex analyses
- Ask Claude to **explain what the code is doing** since you're learning
- Avoid asking for visualizations or plots in these exercises
- Include the output of `glimpse()` in your prompt so Claude knows your variable names

Example prompt: "Using tidyverse in R, calculate the treatment effect by comparing mean post_earnings_12m between the treated and control groups. Also show me how to check compliance by comparing treated vs takeup. Keep the code simple and explain what each line does. Here is what my data looks like: [paste glimpse output]"

3.1 1. Job Training Experiment Analysis

Dataset: data/job_training_rct.csv

Description: Randomized controlled trial evaluating a job training program's effectiveness on employment and earnings outcomes. Participants were randomly assigned to receive job training or serve as controls.

Variables:

- `participant_id`: Unique identifier (integer)
- `age`: Participant age in years (integer)
- `gender`: Gender identity (character: "male", "female", "nonbinary")
- `education`: Education level (integer: 1=Less than HS, 2=HS, 3=Some college, 4=College+)
- `prior_earnings`: Annual earnings before program (integer, dollars)
- `treated`: Treatment assignment (integer: 0=control, 1=treatment)
- `takeup`: Actually participated in training (integer: 0=no, 1=yes)
- `post_earnings_6m`: Earnings 6 months after program (integer, dollars)
- `post_earnings_12m`: Earnings 12 months after program (integer, dollars)
- `employment_12m`: Employed at 12 months (integer: 0=no, 1=yes)
- `center_id`: Training center location (integer)

```r
# Load the dataset
job_training <- read_csv("data/job_training_rct.csv")
```

```
Rows: 500 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (1): gender
dbl (10): participant_id, age, education, prior_earnings, treated, takeup, p...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Explore the data
glimpse(job_training)
```

```
Rows: 500
Columns: 11
$ participant_id    <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 1~
$ age               <dbl> 41, 54, 44, 30, 52, 35, 50, 36, 51, 28, 43, 37, 35, ~
$ gender            <chr> "male", "nonbinary", "male", "male", "female", "male~
$ education         <dbl> 1, 1, 4, 1, 3, 4, 3, 2, 3, 2, 4, 4, 4, 3, 3, 4, 3, 3~
$ prior_earnings    <dbl> 6929, 10965, 19520, 49727, 34381, 23890, 26846, 2610~
$ treated           <dbl> 0, 1, 1, 0, 0, 0, 0, 1, 1, 1, 0, 1, 1, 0, 0, 0, 0, 1~
$ takeup            <dbl> 0, 1, 1, 0, 0, 0, 0, 1, 1, 0, 0, 1, 1, 0, 0, 0, 0, 1~
$ post_earnings_6m  <dbl> 6044, 17438, 21134, 50505, 34871, 29558, 29672, 4040~
$ post_earnings_12m <dbl> 13560, 12104, 20609, 56509, 46957, 26493, 17539, 256~
$ employment_12m    <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
$ center_id         <dbl> 2, 1, 6, 1, 1, 1, 8, 2, 4, 7, 4, 8, 7, 6, 10, 5, 6, ~
```

a. **Group Means**: Use Claude to compute the mean 12-month earnings (`post_earnings_12m`) for the treatment group and, separately, for the control group. Just report each group's average — you don't need to combine them into a single number.

```
group_means <- job_training %>%
group_by(treated) %>%
summarize(
n = n(),
mean_earnings_12m = mean(post_earnings_12m),
sd_earnings_12m = sd(post_earnings_12m)
)
print(group_means)
```

```
# A tibble: 2 x 4
  treated     n mean_earnings_12m sd_earnings_12m
```

```
    <dbl> <int>              <dbl>            <dbl>
1       0   246             30133.           12906.
2       1   254             35251.           12618.
```

b. **Compliance Analysis**: Ask Claude to help you examine the difference between treatment assignment (`treated`) and actual participation (`takeup`). What does this tell us about the experiment?

```r
compliance_table <- job_training %>%
group_by(treated, takeup) %>%
summarize(n = n(), .groups = "drop") %>%
pivot_wider(names_from = takeup, values_from = n, values_fill = 0) %>%
rename(No_Takeup = `0`, Takeup = `1`)
print(compliance_table)
```

```
# A tibble: 2 x 3
  treated No_Takeup Takeup
    <dbl>     <int>  <int>
1       0       242      4
2       1        21    233
```

3.2 2. Civic Crowdfunding Experiment

Dataset: data/civic_crowdfund.csv

Description: Experiment testing whether government matching funds increase success of civic crowdfunding projects. Projects were randomly assigned to receive government matching or not.

Variables:

- `project_id`: Unique project identifier (integer)
- `city_id`: City where project located (integer)
- `goal_amount`: Fundraising goal in dollars (integer)
- `deadline_days`: Days to reach goal (integer)
- `category`: Project type (character: "Transit", "Parks", "Schools", "Arts", "Safety")
- `gov_match`: Government matching treatment (integer: 0=no match, 1=match offered)
- `raised_amount`: Total amount raised (integer, dollars)
- `n_donors`: Number of individual donors (integer)

```r
# Load the dataset
civic_crowd <- read_csv("data/civic_crowdfund.csv")
```

```
Rows: 300 Columns: 8
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (1): category
dbl (7): project_id, city_id, goal_amount, deadline_days, gov_match, raised_...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Explore the data
glimpse(civic_crowd)
```

```
Rows: 300
Columns: 8
$ project_id    <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 1~
$ city_id       <dbl> 43, 33, 14, 7, 30, 43, 14, 35, 42, 38, 40, 8, 13, 2, 6, ~
$ goal_amount   <dbl> 15000, 20000, 50000, 50000, 10000, 50000, 20000, 5000, 5~
$ deadline_days <dbl> 53, 39, 57, 69, 75, 65, 44, 80, 49, 35, 79, 80, 80, 83, ~
$ category      <chr> "Safety", "Transit", "Transit", "Transit", "Parks", "Sch~
$ gov_match     <dbl> 0, 1, 1, 1, 0, 0, 0, 1, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 0,~
$ raised_amount <dbl> 9927, 28986, 60153, 66741, 9432, 40155, 23438, 4283, 514~
$ n_donors      <dbl> 278, 317, 820, 971, 263, 1285, 304, 80, 1003, 139, 578, ~
```

a. **Success Rate Analysis**: Work with Claude to calculate what percentage of projects reached their funding goals in each treatment condition. Create a success indicator variable first.

An **indicator variable** (also called a dummy variable) is coded `1` when a condition is true and `0` when it is false — here, `1` if a project reached its goal and `0` if it did not. Because R treats `1`/`0` (and `TRUE`/`FALSE`) as numbers, `sum()` of an indicator counts the cases coded 1 and `mean()` of an indicator gives the proportion coded 1 (here, the success rate).

```
civic_crowd <- civic_crowd %>%
mutate(success = if_else(raised_amount >= goal_amount, 1, 0))
# Calculate success rates by treatment
success_rates <- civic_crowd %>%
group_by(gov_match) %>%
summarize(
n_projects = n(),
n_successful = sum(success),
success_rate = mean(success)
)
print(success_rates)
```

```
# A tibble: 2 x 4
  gov_match n_projects n_successful success_rate
      <dbl>      <int>        <dbl>        <dbl>
1         0        145           36        0.248
2         1        155           84        0.542
```

b. **Heterogeneous Effects**: Ask Claude to help you test whether the government matching effect varies by project category. Which types of projects benefit most from matching?

```
category_effects <- civic_crowd %>%
group_by(category, gov_match) %>%
summarize(success_rate = mean(success), .groups = "drop") %>%
pivot_wider(names_from = gov_match, values_from = success_rate,
names_prefix = "match_") %>%
mutate(treatment_effect = match_1 - match_0) %>%
arrange(desc(treatment_effect))
print(category_effects)
```

```
# A tibble: 5 x 4
  category match_0 match_1 treatment_effect
  <chr>      <dbl>   <dbl>            <dbl>
1 Safety     0.1     0.625           0.525
2 Transit    0.194   0.545           0.351
3 Schools    0.24    0.579           0.339
4 Arts       0.292   0.462           0.170
5 Parks      0.433   0.5             0.0667
```

3.3 3. Refugee Integration Study

Dataset: data/refugee_integration.csv

Description: Longitudinal study examining factors affecting refugee integration outcomes, including the effect of English language training hours on employment and citizenship application.

Variables:

- `case_id`: Unique case identifier (integer)
- `arrival_year`: Year of arrival in country (integer)
- `country_origin`: Origin country code (character)
- `family_size`: Number of family members (integer)
- `english_class_hours`: Hours of English instruction received (integer)
- `employed_year1`: Employed in first year (integer: 0=no, 1=yes)

- `income_year5`: Income in fifth year, dollars (integer)
- `citizenship_applied`: Applied for citizenship (integer: 0=no, 1=yes)

```r
# Load the dataset
refugee_data <- read_csv("data/refugee_integration.csv")
```

```
Rows: 400 Columns: 8
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (1): country_origin
dbl (7): case_id, arrival_year, family_size, english_class_hours, employed_y...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Explore the data
glimpse(refugee_data)
```

```
Rows: 400
Columns: 8
$ case_id             <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15,~
$ arrival_year        <dbl> 2022, 2022, 2022, 2022, 2015, 2020, 2022, 2020, 20~
$ country_origin      <chr> "AF", "VE", "ET", "VE", "MM", "SY", "AF", "MM", "A~
$ family_size         <dbl> 7, 5, 3, 8, 7, 4, 4, 5, 8, 6, 5, 7, 1, 7, 1, 6, 3,~
$ english_class_hours <dbl> 413, 340, 288, 135, 477, 361, 475, 387, 489, 356, ~
$ employed_year1      <dbl> 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 0, 1, 1, 1,~
$ income_year5        <dbl> 37516, 38191, 9234, 31884, 54854, 38582, 41537, 45~
$ citizenship_applied <dbl> 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0,~
```

a. **Language Training Effect**: Use Claude to analyze whether English class hours predict employment in the first year. What's the relationship between language training and employment outcomes?

```r
refugee_data <- refugee_data %>%
mutate(
training_level = case_when(
english_class_hours == 0 ~ "None",
english_class_hours < 150 ~ "Low",
english_class_hours < 300 ~ "Medium",
TRUE ~ "High"
),
```

```r
training_level = factor(training_level,
levels = c("None", "Low", "Medium", "High"))
)
# Employment rates by training level
employment_by_training <- refugee_data %>%
group_by(training_level) %>%
summarize(
n = n(),
employment_rate = mean(employed_year1),
.groups = "drop"
)
print(employment_by_training)
```

```
# A tibble: 3 x 3
  training_level     n employment_rate
  <fct>          <int>           <dbl>
1 Low              111           0.252
2 Medium           123           0.683
3 High             166           0.675
```

b. **Long-term Outcomes**: Ask Claude to help you examine how arrival year affects long-term integration (5-year income and citizenship applications). Are there cohort effects?

```r
cohort_outcomes <- refugee_data %>%
group_by(arrival_year) %>%
summarize(
n = n(),
pct_employed_year1 = mean(employed_year1) * 100,
mean_income_year5 = mean(income_year5, na.rm = TRUE),
n_with_income = sum(!is.na(income_year5)),
pct_citizenship = mean(citizenship_applied) * 100,
.groups = "drop"
)
print(cohort_outcomes)
```

```
# A tibble: 9 x 6
  arrival_year     n pct_employed_year1 mean_income_year5 n_with_income
         <dbl> <int>              <dbl>             <dbl>         <int>
1         2015    51               43.1            34550.            51
2         2016    43               62.8            33077.            43
3         2017    48               43.8            30217.            48
```

```
4          2018    37                48.6            31302.            37
5          2019    53                64.2            31192.            53
6          2020    46                56.5            31289.            46
7          2021    46                56.5            28110.            46
8          2022    39                71.8            31458.            39
9          2023    37                59.5            26813.            37
# i 1 more variable: pct_citizenship <dbl>
```

3.4 4. Climate Anxiety Panel Study

Dataset: data/climate_anxiety.csv

Description: Panel study tracking climate anxiety levels and behavioral responses over time across different political ideologies and news consumption patterns.

Variables:

- `respondent_id`: Unique respondent identifier (integer)
- `wave`: Survey wave number (integer: 1, 2, 3)
- `age`: Respondent age (integer)
- `political_ideology`: Political orientation (integer: 1=Very liberal, 2=Liberal, 3=Moderate, 4=Conservative, 5=Very conservative)
- `climate_anxiety_scale`: Climate anxiety score 0-15 (integer, higher=more anxious)
- `behavior_change_index`: Pro-environmental behavior changes 0-10 (integer, higher=more changes)
- `news_consumption_freq`: Climate news frequency (integer: 1=Never, 2=Rarely, 3=Sometimes, 4=Often, 5=Daily)

```r
# Load the dataset
climate_data <- read_csv("data/climate_anxiety.csv")
```

```
Rows: 600 Columns: 7
-- Column specification --------------------------------------------------------
Delimiter: ","
dbl (7): respondent_id, wave, age, political_ideology, news_consumption_freq...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Explore the data
glimpse(climate_data)
```

```
Rows: 600
Columns: 7
$ respondent_id         <dbl> 1, 1, 1, 2, 2, 2, 3, 3, 3, 4, 4, 4, 5, 5, 5, 6, ~
$ wave                  <dbl> 1, 2, 3, 1, 2, 3, 1, 2, 3, 1, 2, 3, 1, 2, 3, 1, ~
$ age                   <dbl> 29, 29, 29, 70, 70, 70, 69, 69, 69, 75, 75, 75, ~
$ political_ideology    <dbl> 5, 5, 5, 5, 5, 5, 1, 1, 1, 3, 3, 3, 2, 2, 2, 4, ~
$ news_consumption_freq <dbl> 4, 4, 4, 3, 3, 3, 5, 5, 5, 2, 2, 2, 1, 1, 1, 4, ~
$ climate_anxiety_scale <dbl> 6, 6, 5, 7, 6, 7, 10, 8, 10, 4, 3, 12, 5, 4, 8, ~
$ behavior_change_index <dbl> 4, 2, 3, 5, 1, 2, 5, 6, 3, 4, 1, 5, 2, 0, 5, 2, ~
```

a. **Ideology and Anxiety**: Work with Claude to examine how political ideology relates to climate anxiety levels. Calculate mean anxiety by ideology group.

```r
climate_data <- climate_data %>%
mutate(
ideology_label = factor(political_ideology,
levels = 1:5,
labels = c("Very Liberal", "Liberal", "Moderate",
"Conservative", "Very Conservative"))
)
# Mean anxiety by ideology
anxiety_by_ideology <- climate_data %>%
group_by(ideology_label) %>%
summarize(
n = n(),
mean_anxiety = mean(climate_anxiety_scale),
sd_anxiety = sd(climate_anxiety_scale),
.groups = "drop"
)
print(anxiety_by_ideology)
```

```
# A tibble: 5 x 4
  ideology_label        n mean_anxiety sd_anxiety
  <fct>             <int>        <dbl>      <dbl>
1 Very Liberal        147         9.44       2.16
2 Liberal             123         8.51       2.01
3 Moderate            102         7.30       2.57
4 Conservative        105         6.72       2.27
5 Very Conservative   123         5.48       2.45
```

b. **Panel Analysis**: Ask Claude to help you analyze how climate anxiety changes over time (across waves) and whether this varies by news consumption frequency.

```r
anxiety_by_wave <- climate_data %>%
group_by(wave) %>%
summarize(
n = n(),
mean_anxiety = mean(climate_anxiety_scale),
.groups = "drop"
)
print(anxiety_by_wave)
```

```
# A tibble: 3 x 3
   wave     n mean_anxiety
  <dbl> <int>        <dbl>
1     1   200         6.93
2     2   200         7.78
3     3   200         8.09
```